

ACTIVE GALAXIES

There are no simple rules governing the appearance of active galaxies. Some have a disrupted structure seen either in visible light or at other wavelengths, while others appear normal at first but radiate unusually large amounts of energy at certain wavelengths. In fact, the majority of galaxies show activity of one kind or another. However, a smaller proportion of galaxies have particularly active nuclei, powered by matter spiraling into their central black hole. These

JET FROM AN ACTIVE GALAXY

Pictured in radio waves and false colors, this jet of particles blasted from the core of the galaxy M87 is a typical feature of active galaxies with black-hole engines.

include Seyfert galaxies, radio galaxies, quasars, and blazars. The vast majority of known active galaxies are distant quasars. Objects lying nearer to the Milky Way, although less spectacularly violent, are at least close enough for astronomers to study in detail.

TYPE-II SEYFERT GALAXY

Circinus Galaxy



CATALOG NUMBER
ESO 97-G13

SHAPE Sb spiral

DISTANCE
13 million light-years

DIAMETER
37,000 light-years

MAGNITUDE 11.0

CIRCINUS

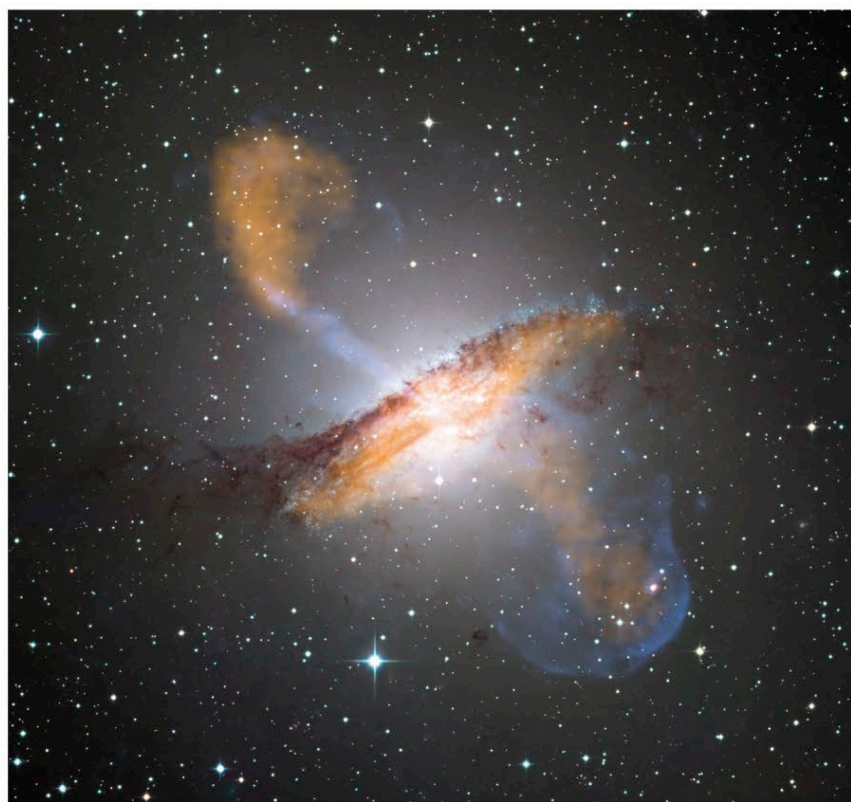
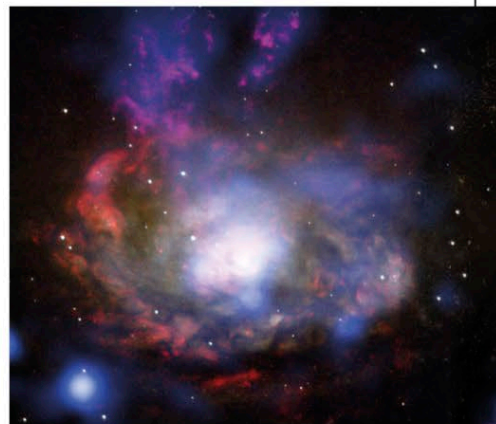
Although it is one of the nearest active galaxies to Earth, the spiral galaxy in Circinus went undiscovered until just a few decades ago. It remained hidden

for so long partly because it lies just 4 degrees below the plane of the Milky Way and is obscured by star clouds. The full extent of the Circinus Galaxy's extraordinary nature was revealed only when it was observed by the Hubble Space Telescope in 1999. The galaxy is a Seyfert (see p.320)—a spiral with an unusually bright, compact region at its core, thought to result from material slowly drifting onto a massive central black hole. Hubble's infrared camera revealed how the galaxy's gas is concentrated in a central ring, just 250 light-years in diameter, around the black hole. Also apparent is a loose outer ring in the plane of

the galaxy, around 1,300 light-years across, where great bursts of star formation are occurring. Finally, Hubble showed a cone-shaped cloud billowing above the plane of the galaxy. This is matter ejected by the magnetic fields of the black hole and glows as it is heated by the ultraviolet radiation from the nucleus.

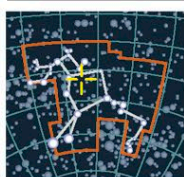
CONE OF MATTER

The pinkish-white region near the core of the Circinus Galaxy shows where matter is being flung out, in a cone shape, from the central black hole into the gas cloud above the galaxy.



RADIO GALAXY

Centaurus A



CATALOG NUMBER
NGC 5128

SHAPE Peculiar elliptical

DISTANCE
15 million light-years

DIAMETER
80,000 light-years

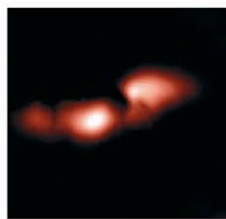
MAGNITUDE 7.0

CENTAURUS

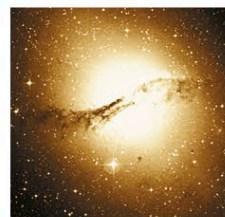
A ball of old yellow stars, NGC 5128 shows some features typical of an elliptical galaxy, but its most striking aspect is the dark dust lane that cuts across it, bisecting the uniform glow of stars with a ragged silhouette. What is more, the galaxy is at the center of a pair of vast radio lobes, 1 million light-years across. The name of this radio source, Centaurus A, is now the most widely used name for the galaxy itself. Astronomers have studied Centaurus A in detail at a range of



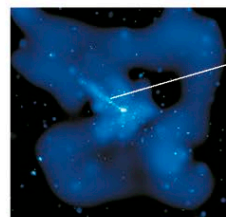
RADIO CONTINUUM



RADIO (21-CM WAVELENGTH)



OPTICAL WAVELENGTHS



X-RAY WAVELENGTHS

COMPOSITE VIEW

Centaurus A has been imaged at various wavelengths (left and below). The image at far left is a composite at optical, microwave, and X-ray wavelengths.

DUSTY DISK

This Hubble Space Telescope close-up of Centaurus A (right) reveals dark interstellar dust, glowing orange gas clouds, and brilliant blue star clusters formed in the collision between two galaxies.

wavelengths. The Hubble Space Telescope looked through the dust lanes

with its infrared camera and found a huge accretion disk at the center—a sure sign of an active black hole pulling in matter at Centaurus A's core.

It is now generally agreed that NGC 5128 is an elliptical galaxy absorbing a spiral. The ghost of the spiral is shown by the dust lane and by the bright star clusters that stud it—perhaps generated by shock waves as the two galaxies merge.

